

Practical No.4

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Objective:

A random sample is taken from $B(5, p)$ to test

$H_0 : p = 0.5$ against

$H_A : p = 0.7$.

It is decided to reject H_0 : if $x \geq 3$.

find

a) α

Python Code

```
In [1]: import math
def binomial(n,p,k):
    tp=0
    for x in range(k,n+1):
        prob = (math.factorial(n)/(math.factorial(x)*math.factorial(n-x))) * (p**x) * ((1-p)**(n-x))
        tp += prob
    return tp
```

```
In [2]: binomial(5,0.5,3)
```

```
Out[2]: 0.5
```

Interpretation:

As we can see the $\alpha = P(\text{Reject } H_0 \mid H_0) = 0.5$ by using own 'binomial' function. So we reject H_0 as per test with $\alpha = 0.5$